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Human induced pluripotent stem cell-derived cardiomyocytes to study inflammation-induced diastolic dysfunction

Yuki Tatekoshi, Chunlei Chen, Jason Shapiro, Hsiang-Chun Chang, Malorie Blancard, Paul W. Burridge, Paul Burridge, Matthew Feinstein, Richard D'Aquila, Priscilla Hsue, Hossein Ardehali 

Feinberg Cardiovascular and Renal Research Institute, and Northwestern University, Chicago, IL • Department of Pharmacology, Northwestern University, Chicago, IL • Department of Medicine, Northwestern University, Chicago, IL • Department of Medicine, University of California at San Francisco, San Francisco, CA

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Abstract

Heart failure with preserved ejection fraction (HFpEF) is commonly found in persons living with HIV (PLWH) even when antiretroviral therapy (ART) suppresses HIV viremia. However, studying this condition has been challenging because an appropriate animal model is not available. In this paper, we study the relaxation function of human induced pluripotent stem cells-derived cardiomyocytes (hiPSC-CMs) in culture to simulate HFpEF and to assess whether various drugs have an effect. We show that treatment of hiPSC-CMs with inflammatory cytokines (such as interferon- γ or TNF- α) impair their relaxation and that SGLT2 inhibitors, clinically proven as effective for HFpEF, reverse this effect.

Additionally, treatment with mitochondrial antioxidants (like mito-Tempo) and certain antiretrovirals resulted in the reversal of the effects of these cytokines on cardiomyocyte relaxation. Finally, incubation of hiPSC-CMs with serum from HIV patients with and without diastolic dysfunction (DD) did not alter their relaxation, indicating that short exposure to the serum of these patients is not sufficient to induce DD *in vitro*. Together, our results indicate that hiPSC-CMs can be used as a model to study molecular mechanisms of inflammation-mediated abnormal cardiomyocyte relaxation and screen for potential new interventions.

eLife assessment

Tatekoshi et al. endeavored to utilize human-induced pluripotent stem cell-derived cardiomyocytes to create an in vitro model of Heart Failure with preserved Ejection Fraction (HFpEF) to investigate the impacts of inflammation, sera from HIV patients, and the application of cardioprotective and antiviral medications on cardiomyocyte relaxation as a proxy for diastolic function. Their investigations revealed that inflammatory cytokines lead to an increased decay in calcium transients, a process that could be alleviated through the use of SGLT2 inhibitors and mitochondrial antioxidants. These results indicate that inflammation might contribute to diastolic dysfunction and suggest that SGLT2 inhibitors and mitochondrial antioxidants could offer cardioprotective benefits by reducing inflammation in cardiomyocytes. These **valuable**, yet **incomplete** results, can be complemented by employing biomechanical and molecular analyses, alongside validation of the therapeutic duration and dosages.

Introduction

Heart failure (HF) with preserved ejection fraction (HFpEF) refers to a form of HF that is associated with diastolic dysfunction (DD). On echocardiography, HFpEF is associated with a greater than 50% of blood in the left ventricle (LV) is pumped with each beat but the LV displays impaired relaxation and increased stiffness during the filling phase of cardiac cycle. Echocardiographic evidence of DD is associated with higher all-cause mortality even in asymptomatic patients (1). The pathophysiology underlying DD is known to be an increase in the stiffness of the LV, although there are limited ways to study mechanisms and interventions. Studies have shown that patients with HFpEF have multiple comorbidities, such as obesity, hypertension, renal dysfunction, and diabetes (2,3) and that it is associated with aging (4). It is also proposed that these comorbid conditions each lead to increased systemic inflammation which then causes myocardial remodeling and fibrosis (2,5–7).

There are few options for the treatment of HFpEF, despite the increasing incidence of this disease in HIV-uninfected and -infected patients. The current treatments relieve symptoms and target comorbid conditions, such as hypertension, diabetes, coronary artery disease, hyperlipidemia, and atrial fibrillation. Although beta blockers (BB), ACE inhibitors (ACEI), angiotensin receptor blockers (ARB), and cardiac resynchronization therapy improve outcome in patients with HF with reduced ejection fraction (HFrEF), no clinical benefits have been shown with any of these drugs in HFpEF. However, they are still empirically used to treat HFpEF patients in clinical practice. The only drugs that have shown clinical benefit in rigorous trials are antagonists of the mineralocorticoid receptor (MRA, such as spironolactone) (8) and inhibitors of the sodium glucose co-transporter 2 (SGLT2, such as empagliflozin) (9).

Persons living with HIV (PLWH) have a higher prevalence of HFpEF compared to uninfected individuals (10,11). Recent studies have shown that LV mass index and DD are each significantly worse in PLWH with HFpEF than among persons living without HIV with HFpEF, despite similar EF. These features were associated with lower nadir CD4⁺ T-cell count, suggesting that this process could be due to the level of immunodeficiency and/or duration of untreated infection (12). A number of studies attempted to determine the mechanism for increased HFpEF in PLWH. Administration of HIV gp120 resulted in DD after adrenergic stimulation in rats (13), and caused negative inotropic effects in adult rat ventricular myocytes (14,15). Macaques infected with simian immunodeficiency virus (SIV) display DD, which was associated with the

degree of myocardial SIV viral load. Inhibition of CCR5, a co-receptor with CD4 for HIV gp120-mediated entry into cells, in the SIV-infected rhesus macaque model of DD preserved cardiac diastolic function (16 [↗](#)).

Untreated HIV infection is associated with circulating inflammatory cytokines and chemokines (17 [↗](#)), and many remain elevated even when HIV viremia is well-controlled by antiretroviral therapy (ART) (18 [↗](#), 19 [↗](#)). For instance, elevated IFN- γ is observed during acute HIV viremia and remained elevated despite highly active ART (HAART) (20 [↗](#)). Higher IFN- γ is also associated with impaired CD4 recovery in HIV after the initiation of HAART (21 [↗](#)). Additionally, while the immune system resets itself after the regression of bacterial or viral infection, it remains hypersensitive when ART suppresses viremia in chronic HIV infection. Of note, monocytes from PLWH treated with viremia suppressing ART demonstrated increased production of TNF- α and IL-6 after stimulation with lipopolysaccharide (LPS) (22 [↗](#)), indicating the presence of immune system rewiring during HIV infection that is not reversed with effective ART. Therefore, cardiomyocytes in PLWH are exposed to higher circulating inflammatory cytokines at baseline despite effective ART, and with further elevation of these inflammatory cytokines after concomitant infections. Among these cytokines, TNF- α is associated with mortality (23 [↗](#)) and severe coronary stenosis (greater than 70% stenosis) (24 [↗](#)) in PLWH.

There are limited studies to address the molecular basis of cardiomyocyte dysfunction in HIV-mediated HFpEF or the effectiveness of various drugs. Additionally, there are currently no models using human heart cells and HIV. These major limitations in HIV-related HFpEF raise the critical need for a novel approach to study this complicated disease. Human induced pluripotent stem cell (hiPSC)-derived cardiomyocytes (hiPSC-CM) represent a novel technology that has been successfully applied to understanding basic mechanisms of several cardiovascular diseases, including long QT (25 [↗](#)), LEOPARD (26 [↗](#)) and Timothy (27 [↗](#)) syndromes, dilated cardiomyopathy (CM) (28 [↗](#), 29 [↗](#)) and hypertrophic CM (29 [↗](#)). They are also used for efficacy and toxicity screening of drugs (30 [↗](#), 31 [↗](#)), leading to enhanced understanding of cellular mechanisms at the single cardiomyocyte level, not previously possible using human cells. Furthermore, hiPSC-CMs have been used to study a model of viral myocarditis (i.e., coxsackie virus) and to predict the efficacy of antiviral drug therapies (32 [↗](#)), indicating that these cells can be used to study HF due to viral infections. In this paper, we show that hiPSC-CMs can be used as a model for DD in response to inflammatory cytokines *in vitro* and to assess the effects of various drugs on their relaxation parameters. Given the major limitations in the field of HIV-mediated HFpEF, this model will allow investigators to study the mechanism of cardiac DD in inflammatory condition *in vitro* and assess the effectiveness of various drugs to reverse impaired relaxation noted with systemic inflammation.

Results

TNF- α and IFN- γ impair relaxation in hiPSC-CMs without causing cellular damage

We first generated human induced pluripotent stem cells (hiPSCs) and differentiated into cardiomyocytes using the protocols previously published by our groups (33 [↗](#)). We then assessed whether treatment with cytokines would have detrimental effects on these cells. Because DD has been associated with mitochondrial dysfunction (34 [↗](#)), we first measure whether cytokine treatment results in changes of mitochondrial membrane potential (MMP) at baseline. For these experiments, cells were treated with cytokines at indicated concentrations for 48 hours. These studies revealed that each of the cytokines studied in our system did not alter MMP in hiPSC-CMs (Figure 1-figure supplement 1 [↗](#)). For the remaining studies, we focused our experiments on two of these cytokines that often are chronically increased in PLWH, TNF- α and IFN- γ .

We next assessed whether treatment with TNF- α or IFN- γ alters diastolic function in hiPSC-CMs by measuring calcium transient and assessing the decay time and downstroke time in these cells using Single Cell Kinetic Image Cytometry (Vala science). Treatment with both TNF- α and IFN- γ resulted in a significant increase in decay time and downstroke time, which are each markers of diastolic function, while beats per minute (indirect measurement of heart rate and not diastolic function) was essentially unchanged (**Figure 1A-C**). Additionally, treatment with mitoTempo (an agent with mitochondrial antioxidant activity) reversed the changes that occurred with TNF- α (**Figure 1A and B**), indicating that a reduction in mitochondrial reactive oxygen species (ROS) may, at least partially, reverse the relaxation defect induced by TNF- α .

Since mitoTempo partially reversed the effects of TNF- α on relaxation of hiPSC-CMs, we next assessed whether TNF- α and IFN- γ have any effects on mitochondrial respiration and whether these potential defects would be reversed with antioxidants. For these experiments, we used Seahorse XF96 Analyzer to measure the oxygen consumption rate (OCR) after treatment with TNF- α or IFN- γ in the presence and absence of the general antioxidant, N-acetylcysteine (NAC). Administration of NAC to iPS-CMs supplemented with TNF- α significantly reduced OCR at baseline, although no alteration was observed after adding CCCP, or Antimycin-A and Rotenone. (**Figure 2**) These data indicate that mitochondrial respiration and metabolism might be associated with observed beneficial effects of anti-oxidant reagents on a decay time and a down stroke time. We next used this platform to study the effects of various drugs on *in vitro* measures of DD.

Certain antiviral drugs and SGLT2 inhibitors reverse impaired relaxation in hiPSC-CMs

Given that we were able to induce impaired relaxation in hiPSC-CMs after treatment with inflammatory markers, we assessed whether various antiretrovirals and other drugs that potentially target the pathogenesis of HFpEF can reverse the progression of DD *in vitro*. We first assessed whether ART drugs at various concentrations cause cell death or increased ROS production in hiPSC-CMs. We treated hiPSC-CMs with tenofovir (a nucleotide-analog reverse transcriptase inhibitor), darunavir (a protease inhibitor), raltegravir and elvitegravir (integrase inhibitors) at concentrations between 3 μ M – 10 mM. The dose range was chosen to extend to 10-fold above the IC50 concentrations and reflects the upper range of circulating drug concentration in patients receiving these medications (35–38). In this dose range, we did not observe apparent changes in cell viability and cellular ROS levels (**Figure 3-figure supplement 1A and B**), indicating that exposure to these single drugs alone has minimal effects on cell survival.

We then assessed whether ART can reverse the impaired relaxation induced by TNF- α . For these studies, we used the following single drug concentrations: 5 μ M tenofovir, 10 μ M darunavir, 3 μ M raltegravir, and 10 μ M emtricitabine (an HIV nucleoside analog reverse transcriptase inhibitor). As shown in **Figure 3**, while TNF- α increased the decay time in hiPSC-CMs, treatment with tenofovir, darunavir, raltegravir, and emtricitabine reversed these effects. These results indicate that ART may have beneficial effects on the inflammation-mediated DD that occurs in HIV patients, raising the possibility that its potential benefits may go beyond their inhibitory effects on the virus itself.

A number of other drugs have also been proposed to potentially exert beneficial effects in HFpEF. Among these, we focused on an oral soluble guanylate cyclase (sGC) stimulator (riociguat, 1 μ M), a phosphodiesterase (PDE)-5 inhibitor (sildenafil, 1 μ M), a PDE9 inhibitors (PF-04447943, 5 μ M) and a sodium-glucose transport (SGLT)-2 inhibitor (dapagliflozin, 10 μ M). The dosages are based on previous functional studies in isolated cardiomyocytes (39–43). While TNF- α caused a significant increase in decay time in hiPSC-CMs, riociguat, sildenafil and PF-04447943 failed to reverse this effect (**Figure 4A-C**), indicating that they do not have the properties to reverse the inflammation-mediated impaired relaxation. However, SGLT2 inhibitor dapagliflozin significantly

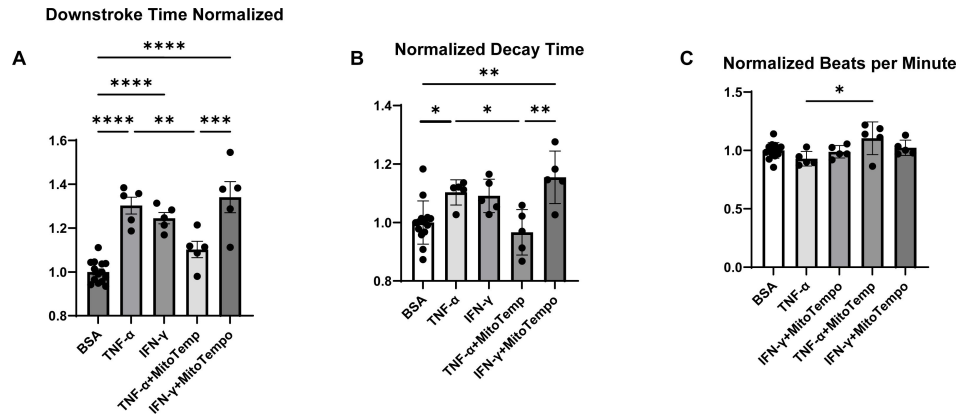


Figure 1.

Normalized downstream time (A), decay time (B), and beating rate as assessed by beats per minute (C) in hiPSC-CM treated with BSA (control), TNF- α , IFN- γ , TNF- α plus mitoTempo, and IFN- γ plus mitoTempo. N=5-14. Data were analyzed by ordinary one-way ANOVA and post-hoc Tukey's multiple comparison test. Bars represent group mean.

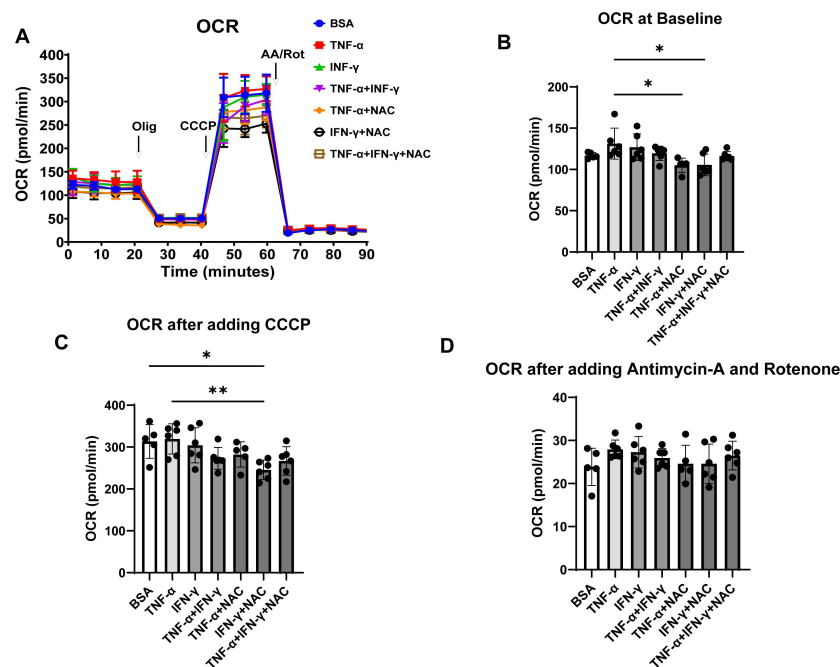


Figure 2.

Oxygen consumption rate (OCR) in hiPSC-CM treated with cytokines and antioxidant agent NAC. (A) OCR trace of hiPSC-CM treated with BSA, TNF- α , IFN- γ , TNF- α +IFN- γ , TNF- α +NAC, IFN- γ +NAC, and TNF- α +IFN- γ +NAC. (B-D) Bar graph summary of data in Panel A with OCR at baseline (B), after adding carbonyl cyanide m-chlorophenyl hydrazone (CCCP) (C), and after adding antimycin A and rotenone (D). N=5-6. Data were analyzed by ordinary one-way ANOVA and post-hoc Tukey's multiple comparison test. Bars represent group mean.

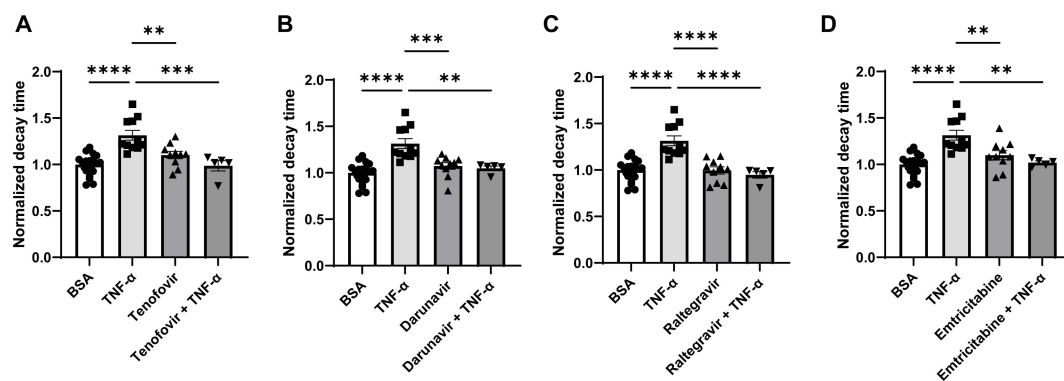


Figure 3.

Decay time (a measurement of cardiomyocyte relaxation) in hiPSC-CM after treatment with TNF-α and various ART drugs. **(A)** Decay time after treatment with TNF-α, tenofovir, and TNF-α+tenofovir. **(B)** Decay time after treatment with TNF-α, darunavir, and TNF-α+darunavir. **(C)** Decay time after treatment with TNF-α, raltegravir, and TNF-α+ raltegravir. **(D)** Decay time after treatment with TNF-α, emtricitabine, and TNF-α+ emtricitabine. N=5-21. Data were analyzed by ordinary one-way ANOVA and post-hoc Tukey's multiple comparison test. Bars represent group mean.

reversed the impaired relaxation induced by TNF- α in hiPSC-CMs (**Figure 4D** [↗](#)). Additionally, we repeated the experiments with IFN- γ and dapagliflozin and showed that the drug is able to reverse the impaired relaxation associated with IFN- γ as well (**Figure 4E** [↗](#)). Finally, we demonstrated that dapagliflozin can reverse the defect in downstroke time with both TNF- α and IFN- γ (**Figure 4F and G** [↗](#)). These studies are in accordance with the clinical studies that have shown beneficial effects of SGLT2 inhibitors in HFpEF and suggest that they may also benefit patients with inflammation-induced HFpEF, like HIV+ patients.

Treatment with serum from patients with HIV does not induce impaired relaxation in hiPSC-CMs

Given that we could simulate impaired relaxation in hiPSC-CMs using inflammatory cytokines, we next assessed whether short-term treatment of these cells with serum from HIV+ patients with evidence of DD would also induce impaired relaxation in these patients. We used two sources of serum in our studies. For the first group, we recruited six men with chronic HIV on ART with undetectable viral loads who underwent cardiac magnetic resonance imaging (CMRI) at the Northwestern Memorial Hospital (NMH) for assessment of myocardial perfusion reserve (MPR). Clinical and MRI-based characteristics of these patients are included in **Supplementary Table 1**. Lower values of myocardial perfusion reserve (MPR) indicate impaired global myocardial blood flow in response to vasodilator stress and are associated with higher incidence of HFpEF and poor prognosis ([44](#) [↗](#), [45](#) [↗](#)). The second group included well-controlled patients with HIV at UCSF with echocardiographic evidence of normal cardiac function or DD.

For both of these groups, we treated hiPSC-CMs with 10% serum from these patients for 48hrs, followed by measurement of calcium transients. Additionally, we treated HUVECs with the serum of these patients for 20 hours and assessed their angiogenic potential. For the first group of patients, we stratify into higher (1.9-3.4; N=3) versus lower (1.0-1.1; N=3) MPR values, which correspond to control and DD, respectively. As shown in **Figure 5A-C** [↗](#), serum from HIV+ patients with DD from group 1 did not result in a significant change in relaxation in hiPSC-CMs. Similar results were obtained after 24-hour incubation. Additionally, serum from these patients did not alter the angiogenic potential of HUVECs (**Figure 5D-G** [↗](#) and **Figure 5-figure supplement 1** [↗](#)). We also assessed the effects of serum from HIV+ patients from group 2. The control group for experiments included serum from HIV patients without evidence of DD. Our data indicate that treatment of hiPSC-CMs and HUVECs with 2% serum of these patients for 3 or 24 hours does not alter diastolic function (**Figure 6A-D** [↗](#)) and angiogenic function (**Figure 6E-I** [↗](#)), respectively. For hiPSC-CMs studies, we exposed the cells to serum for 3 hours and 24 hours and did not observe a difference in decay time with either time point. These results suggest that short-treatment of hiPSC-CMs or HUVECs with serum from patients with DD does not induce changes in diastolic function or their angiogenic potential, respectively.

Discussion

HFpEF is a common disease and its incidence continues to increase. Among risk factors for HFpEF, inflammation is believed to be a common mechanism for the development of the disease. The chronic inflammatory condition associated with treated HIV infection is believed to be responsible for the development of HFpEF in these patients. Despite the high incidence of this disorder among the general population and those with HIV, there are currently very few systems available to study this disorder and to screen for drugs that can target the disease. In this paper, we used hiPSC-CMs as a platform to assess whether DD can be induced in these cells in response to inflammatory cytokines. We demonstrate that TNF- α and IFN- γ can induce DD in these cells, which can partially be reversed with mitochondrially targeted antioxidants. We also demonstrate that the majority of ART drugs and SGLT2 inhibitors can reverse the DD associated with inflammatory cytokines.

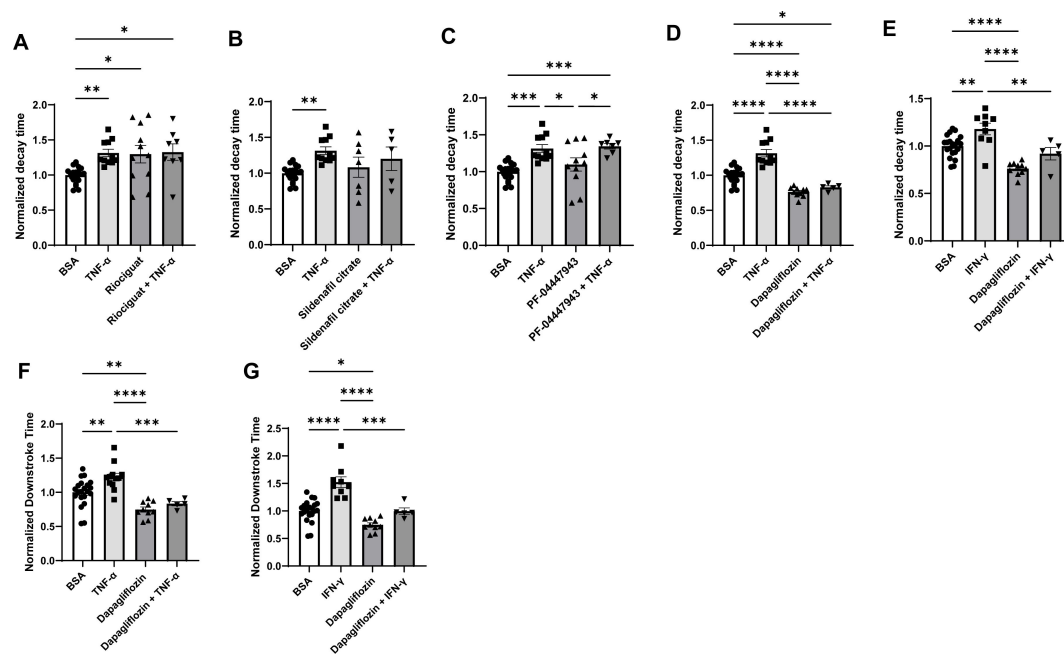


Figure 4.

Decay time in hiPSC-CM after treatment with TNF- α and riociguat, sildenafil citrate, PF-04447943, and dapagliflozin. **(A)** Decay time after treatment with TNF- α , SGS agonist riociguat, and TNF- α +tenofovir. **(B)** Decay time after treatment with TNF- α , sildenafil citrate, and TNF- α +sildenafil citrate. **(C)** Decay time after treatment with TNF- α , PF-04447943, and TNF- α +PF-04447943. **(D)** Decay time after treatment with TNF- α , dapagliflozin, and TNF- α +dapagliflozin. **(E)** Decay time after treatment with IFN- γ , dapagliflozin, and IFN- γ +dapagliflozin. **(F)** Downstroke time after treatment with TNF- α , dapagliflozin, and TNF- α +dapagliflozin. **(G)** Downstroke time after treatment with IFN- γ , dapagliflozin, and IFN- γ +dapagliflozin. (N=5-21. Data were analyzed by ordinary one-way ANOVA and post-hoc Tukey's multiple comparison test. Bars represent group mean.

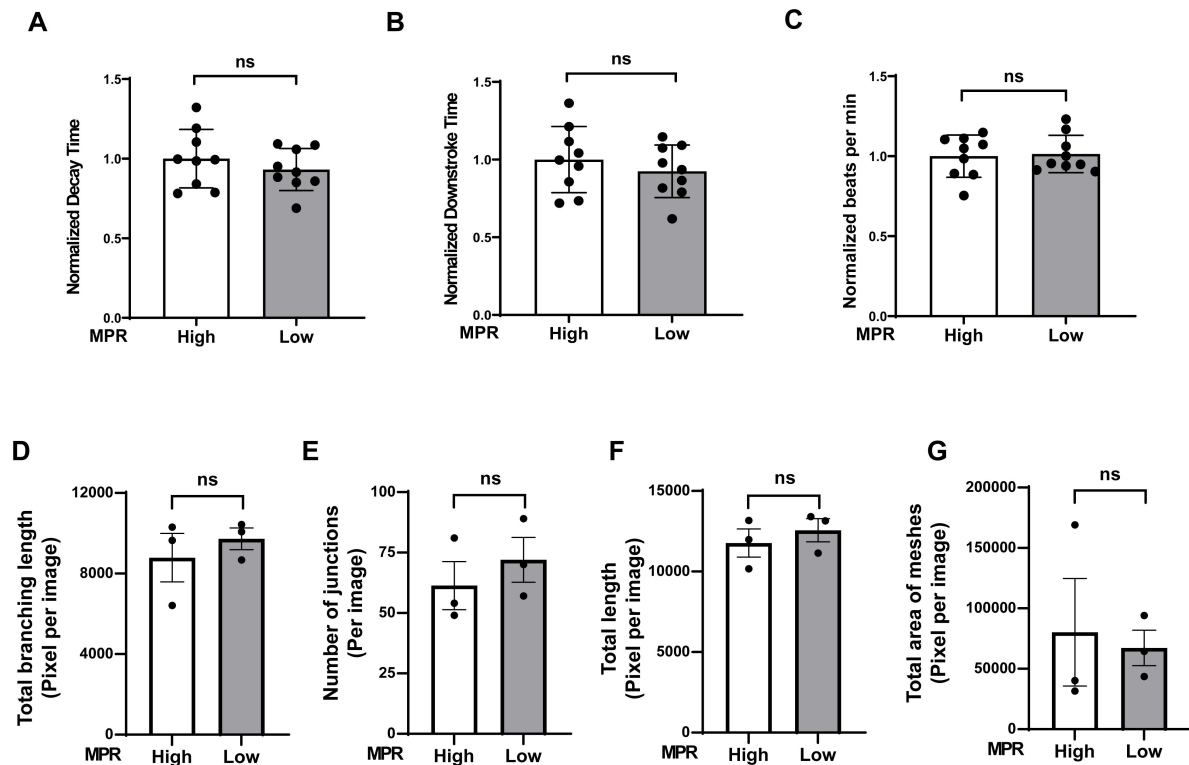


Figure 5.

Diastolic function of hiPSC-CM and angiogenic function of ECs after treatment with 10% serum from Group 1 HIV+ patients with DD. (**A-C**) Pooled normalized decay time (**A**), downstroke time (**B**) and beating rate (**C**) in hiPSC treated with 10% serum for 48 hours from group 1 HIV+ patients with DD. (**D-G**) Angiogenic parameters, including total length of branching (**D**), (N = 3), number of junctions (**E**) (N = 3), total length (**F**) (N = 3) and number of meshes (**G**) (N = 3) in HUVECs after treatment with serum from Group 1 patients for 20 hours. Data were analyzed by unpaired Student's *t* test. Bars represent group mean.

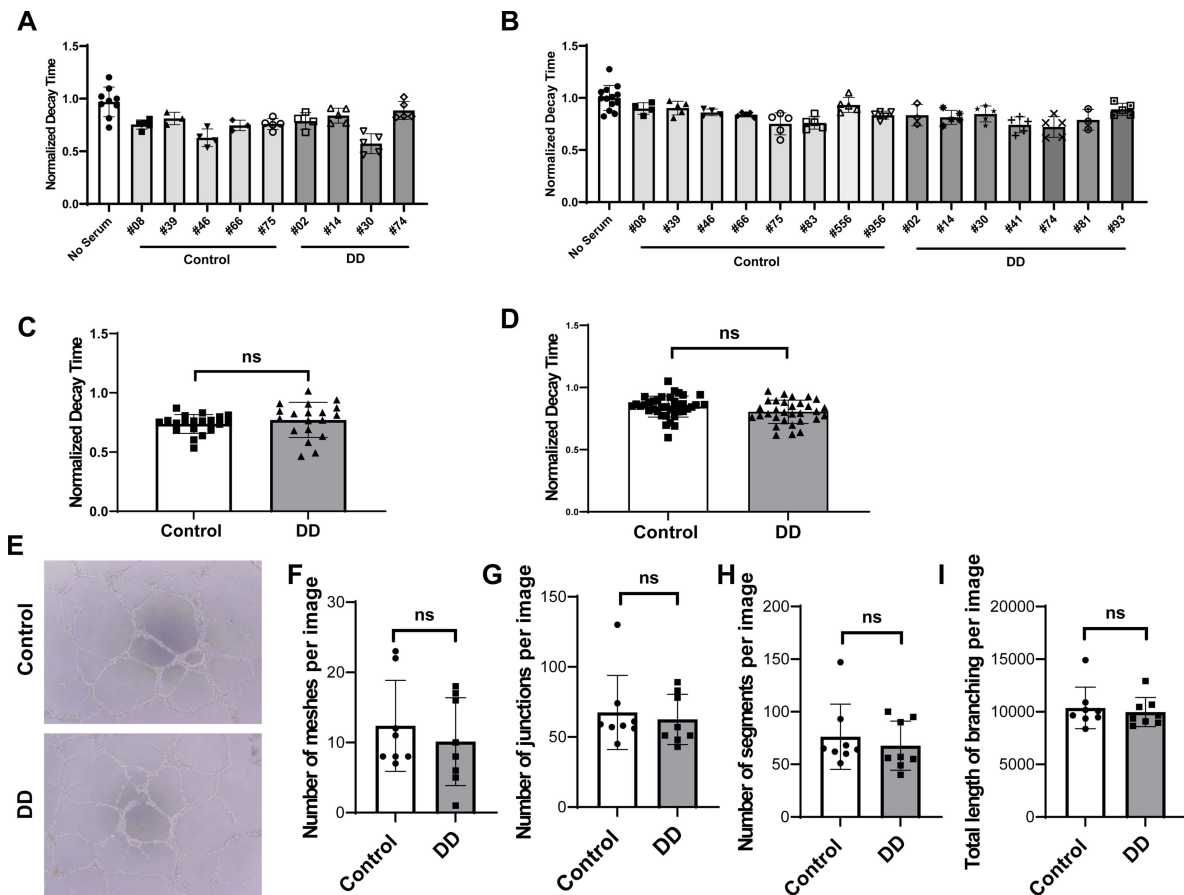


Figure 6.

Diastolic function of hiPSC-CMs and angiogenic function of ECs after treatment with serum from HIV+ patients with DD (Group 2). **(A-B)** Individual normalized decay time 3 hours (N=3-9)**(A)** or 24 hours (N=13)**(B)** after treatment of hiPSC-CM with 2% serum from patients. **(C-D)** Pooled decay time data from patients in panels A and B with 3 hour **(C)** and 24 hours **(D)** after treatment with serum. **(E)** Representative image of formed tube structure with cultured HUVECs treated with serum from HIV+ patients with DD. **(F-I)** Assessment of tube formation of HUVECs 20 hours after treatment with serum of patients, including number of meshes **(F)** (N = 8), number of junctions **(G)** (N = 8), number of segments **(H)** (N = 8) and total length of branching **(I)** (N = 8). Data were analyzed by unpaired Student's *t* test for **(C, D, F, G, H and I)**. Bars represent group mean.

Finally, to determine whether short-term treatment of hiPSC-CMs serum from patients with HIV and DD can also induce DD, we treated these cells with patient serum, but were not able to mimic DD. We also assessed whether serum from these patients can induce defects in angiogenesis, and found that there was no effects on angiogenesis by the serum of these mice. Overall, our results provide a novel platform to study DD associated with inflammatory cytokines and to test the effectiveness of various agents in this disorder.

Studying HFpEF in tissue culture has its own limitations, since HFpEF is a systemic disorder and usually develops as a result of a number of risk factors, including diabetes, hypertension and chronic kidney disease. It is important to note that DD is a major component of HFpEF and studying the relaxation abnormalities associated with this disorder, especially in tissue culture setting, can provide major clues to this disease at the cellular level. Thus, using a cell culture system to mimic the functional abnormality noted in this disorder can provide clues to the disease that can later be tested at the organismal level. Given that there are limited animal models for HFpEF, an *in vitro* model will provide a powerful platform for hypothesis generation and drug screening prior to committing resources for studying these findings at the organismal level.

We demonstrated that treatment with certain cytokines can induce DD in hiPSC-CMs, but serum from HIV patients with imaging evidence of DD did not cause similar abnormalities. This could be due to a number of reasons, including short duration of the treatment with human serum, lower concentration of cytokines in the serum compared to the concentration we used in our studies, and possibly inhibitory effects of other factors in the serum. Nevertheless, it is important to note that cytokines alone can induce DD in hiPSC-CMs and can be used to induce this condition *in vitro*. Further studies are needed to determine whether other conditions can be used to induce DD using serum from patients with DD.

It is not quite surprising that SGLT2 inhibitors reversed the progression of DD in our system, since these drugs have been shown to exert therapeutic effects in HFpEF (9). However, lack of an effect by other drugs we used (including sGC activators and PDE-5 and -9 inhibitors) was unexpected since they have been proposed to have various effects on the vasculature and may benefit other forms of HF. Additionally, the fact that ART drugs were able to reverse the DD associated with cytokine treatment suggest further study of whether these drugs can prevent damage to cardiomyocyte function as a result of the inflammation associated with HIV, in addition to their anti-retroviral effects.

There are limitations with our studies. Darunavir is generally used in combination with ritonavir and elvitegravir along with cobicistat in clinical practice. These additional drugs were not tested in our studies. Additionally, our studies employed only single drugs, which is not commonly used in the clinic. Also, we did not perform a dose-response curve with the drugs we tested. Another limitation of our study is that we did not determine the mechanism of how ART drugs with such different effects on HIV proteins, and minimal effects on any known cellular protein, can have an effect on cellular relaxation. This is an important issue and further studies are needed to better assess the mechanism of how ART can exert such effects.

In summary, we provide a novel approach to study inflammation-induced DD at the cellular level using hiPSC-CMs. We use this method to test a number of drugs and whether they can reverse the DD associated with inflammation. This system can prove useful in studying the molecular basis and potential treatments for HFpEF induced by chronic inflammation.

Supplemental Figure Legends

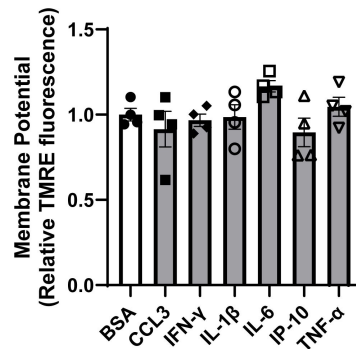


Figure 1-figure supplement 1.

Mitochondrial membrane potential as assessed by tetramethylrhodamine, ethyl ester (TMRE) (A) in hiPSC-CM treated with various cytokines. Concentration of cytokines used is as follows: CCL3 1 nM, CCL4 300 pM, CCL5 10 nM, TNF- α 300 pg/ml, IL-6 1 ng/ml, IL-1 β 200 pg/ml, IP-10 (or CXCL10) 500 pg/ml, IFN- γ 1 ng/ml. N=4. Data were analyzed by ordinary one-way ANOVA and post-hoc Tukey's multiple comparison test. Bars represent group mean.

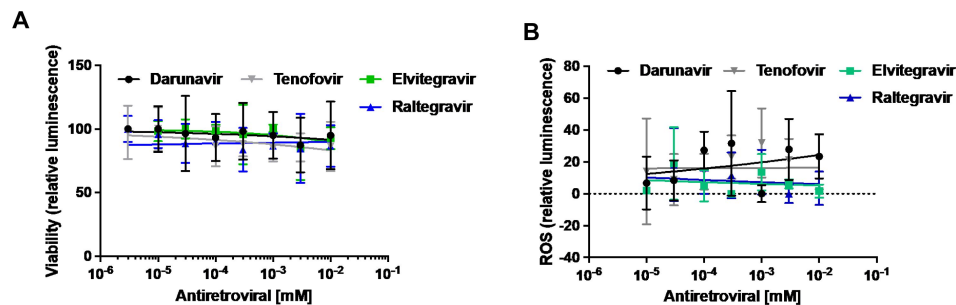


Figure 3-figure supplement 1.

Cell viability (A) and cellular ROS levels (B) in hiPSC-CM exposed to concentrations ranging from between 3nM and 10 μ M of tenofovir (a nucleotide-analog reverse transcriptase inhibitor), darunavir (a protease inhibitor), raltegravir and elvitegravir (integrase inhibitors).

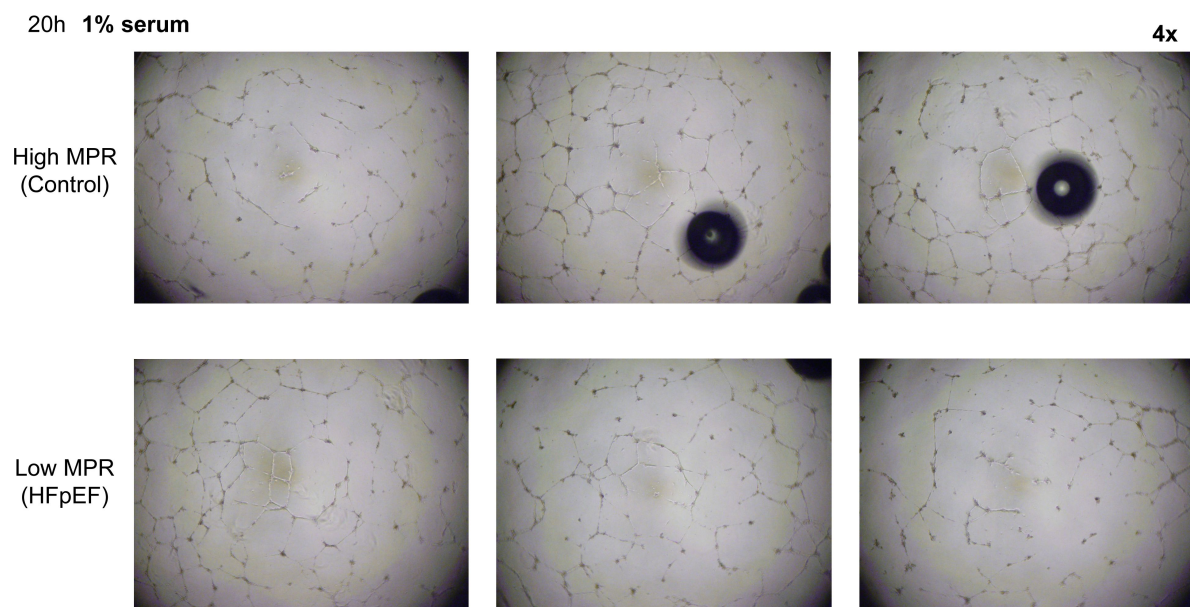


Figure 5-figure supplement 1.

Representative images of formed tube structure with HUVECs treated with 1% serum from HIV+ patients with high or low MPR for 20 hours, or with 1% serum from the same patients for 20 hours.

Methods

Human Induced Pluripotent Cell

Derivation and Cardiac differentiation

Protocols and consents were approved by the institutional review board of Northwestern University. The hiPSC from healthy individuals was generated following the protocol published before (46 [↗](#)). The hiPSC were maintained and differentiated into beating cardiomyocytes, according to the previously published protocol (33 [↗](#)).

TMRE colocalization

Media was removed from hiPSC-CM and were washed with PBS and FluoroBrite DMEM (No Glutamine, No HEPES, No Phenol Red, No Sodium Pyruvate, Thermo A1896702) was added. Cells were stained with 5nM TMRE (for mitochondrial stain) for 20min and then washed with PBS before immunofluorescence. TMRE (red channel) signal was analyzed using ImageJ (FIJI).

Seahorse assay

7 days before assay, hiPSC-CMs were plated at 100,000 cells per well in Matrigel coated Seahorse XF cell culture 96-well plate. The day before the assay, the Seahorse cartridge was placed in the XF calibrant and incubated overnight at 37 °C. On the day of the assay, the plates were incubated at RT for 1hr in glucose free complete DMEM or RPMI without bicarbonate or phenol-red to allow even distribution of cells across the well floor. Before placing the sample plates in the Seahorse XF96 Analyzer, media volume was adjusted to 150 µl in each well. 25 mM glucose, 1.5 mM Oligomycin, 1 µM CCCP, and 20 µM Rot/AA, were diluted in DMEM and injected sequentially into each well, following the standard Seahorse protocol. Cytokine treatment was started 48 hours before Seahorse assay.

Treatment of hiPSC with IFN-γ and TNF-α

hiPSC-CMs after 21-24 days culturing were seeded into 6 wells or 96 wells cell culture plates. After 5 days, IFN-γ at 1ng/ml and TNF-α at 300 pg/ml were added to the cells for 48 hrs.

Drug concentrations

We used the following drug concentrations for our studies: ricociguat - 1 µM, sildenafil citrate - 1 µM, PF-04447943 - 5 µM, atorvastatin - 10 µM, dapagliflozin - 1 µM, tenofovir - 5 µM, emtricitabine - 10 µM, darunavir - 10 µM, raltegravir - 3 µM.

Calcium transient studies in hiPSC-CM

For Ca²⁺ transient analysis, Day 21-24 hiPSC-CMs were plated on the Matrigel coated 96 well black cell culture plate, at a density of 75,000 cells/well. Calcium imaging was performed as described previously (47 [↗](#)). On the day of imaging (Day 28-32), cells were stained with a buffer containing 5 µM Fluo-4AM, NucBlue (2 drops/10ml), 2.5 mM probenecid, 0.02% Pluronic F-127 in FluoroBrite DMEM for 1h at 37 °C, 5% CO₂. After 1h incubation, aspirate dye loading solution and add 200 µl FluoroBrite DMEM to each well for imaging. Single Cell Kinetic Image Cytometry (vala science) was used to scan the plate and CyteSeer Automated Video Analysis Software (vala science) was used to analyze the Ca²⁺ transient.

Tube formation assay

The formation of tube networks was assessed as described before (48 [↗](#)). HUVECs (Lonza) were seeded at 20,000 per well in a 96-well plate coated with 75 mL Matrigel (Fisher Scientific) or Cultrex (R & D Systems) reduced growth factor basement membrane matrix. The cells were treated

with EBM-2 medium containing 1% serum from HIV+ patients obtained from NMH and UCSF wherever mentioned. Following an 20 hour-incubation, tube networks from each biological replicate were analyzed in at least three random fields by light microscopy. The number of branch points (junctions), segments and meshes, and total length of tubule networks (total length of branching) were quantified by Fiji software (Angiogenesis Analyzer).

Participant samples

Blood samples from 6 men with chronic HIV on antiretroviral therapy (ART) with undetectable viral loads who underwent cardiac magnetic resonance imaging (CMRI) for assessment of myocardial perfusion reserve (MPR) were used. For CMRI, all scanning was performed on 1.5T MRI scanners (Siemens Medical Systems), with participants receiving FDA-approved intravenous double-dosing of gadolinium contrast followed by adenosine as the vasodilator agent. Images were performed at maximal hyperemia (vasodilation) and compared with rest: MPR was calculated as the ratio of stress myocardial blood flow to rest myocardial blood flow as described previously (49). Comparisons in iPSC-derived cardiomyocyte performance characteristics were made between those with higher (1.9-3.4; N=3) versus lower (1.0-1.1; N=3) MPR values; additional clinical and MRI-based characteristics are included in **Supplementary Table 1**. The study was approved by the Institutional Review Board of Northwestern University (STU00204874).

Serum samples from UCSF are from an IRB-approved (IRB# 10-03112) longitudinal cohort study evaluating the pathogenesis of HIV and HHV-8 to pulmonary hypertension. PLWH underwent serial phlebotomy, transthoracic echocardiogram, flow-mediated dilation, 6-minute walk test and pulse wave velocity measurements. Subset of participants underwent a right heart catheterization. The study is IRB-approved, and participants provided signed informed consent prior to undergoing study procedures. Serum samples are derived from documented HIV-positive individuals on stable ART and have diastolic function assessed by echocardiogram. Samples are de-identified and stored long term with UCSF's AIDS Specimen Bank (ASB).

Of the 16 participants included from UCSF, median age was 54.1 (IQR 45.0, 57.1), median duration of HIV infection was 17 years (IQR 12.7, 22.7), median CD4 count was 395 (IQR 334, 645), 88% were male, 50% were Caucasian, 37% were African American, and 13% were Hispanic/Latino. Comorbidities included 43.8% had hypertension, 31.3% had hyperlipidemia, 12.5% had diabetes, and 62.5% were current or past smokers. All specimens were matched case-control to those with DD to those without DD.

Baseline echocardiogram characteristics included everyone with preserved LV systolic function ($\geq 50\%$). In the DD group, n=2 had grade 1 DD, n=3 had grade 2 DD, and n=3 had grade 3 DD. Two individuals in DD group had mild left atrial enlargement (LA volume index 35-41 mL/m²). Five individuals in the DD group and two in the control group had RVSP ≥ 35 mmHg.

12 participants (6 from each group) underwent right heart catheterization with two in the diastolic dysfunction group had elevated mPAP (≥ 25 mmHg). Three participants in the DD group and two in the control group had right atrial pressure > 5 mmHg.

Statistical analysis

Data are presented as mean $\pm$ SEM. Unpaired Student t-test and one-way ANOVA, with post hoc Tukey's test were conducted to assess statistical significant difference among experimental groups. A P-value less than 0.05 was considered statistically significant.

References

1. Kardys I., Deckers J. W., Stricker B. H., Vletter W. B., Hofman A., Witteman J. C (2009) **Echocardiographic parameters and all-cause mortality: the Rotterdam Study** *International journal of cardiology* **133**:198–204
2. Dhingra A., Garg A., Kaur S., Chopra S., Batra J. S., Pandey A., Chaanine A. H., Agarwal S. K (2014) **Epidemiology of heart failure with preserved ejection fraction** *Current heart failure reports* **11**:354–365
3. Haass M., Kitzman D. W., Anand I. S., Miller A., Zile M. R., Massie B. M., Carson P. E (2011) **Body mass index and adverse cardiovascular outcomes in heart failure patients with preserved ejection fraction: results from the Irbesartan in Heart Failure with Preserved Ejection Fraction (I-PRESERVE) trial** *Circulation. Heart failure* **4**:324–331
4. Xu B., Daimon M (2016) **Cardiac aging phenomenon and its clinical features by echocardiography** *Journal of echocardiography* **14**:139–145
5. Cheng J. M., Akkerhuis K. M., Battes L. C., van Vark L. C., Hillege H. L., Paulus W. J., Boersma E., Kardys I. (2013) **Biomarkers of heart failure with normal ejection fraction: a systematic review** *European journal of heart failure* **15**:1350–1362
6. Sanders-van Wijk S., van Empel V., Davarzani N., Maeder M. T., Handschin R., Pfisterer M. E., Brunner-La Rocca H. P., investigators, T.-C. (2015) **Circulating biomarkers of distinct pathophysiological pathways in heart failure with preserved vs. reduced left ventricular ejection fraction** *European journal of heart failure* **17**:1006–1014
7. Santhanakrishnan R. *et al.* (2012) **Growth differentiation factor 15, ST2, high-sensitivity troponin T, and N-terminal pro brain natriuretic peptide in heart failure with preserved vs. reduced ejection fraction** *European journal of heart failure* **14**:1338–1347
8. Pfeffer M. A., Braunwald E (2016) **Treatment of Heart Failure With Preserved Ejection Fraction: Reflections on Its Treatment With an Aldosterone Antagonist** *JAMA cardiology* **1**:7–8
9. Anker S. D. *et al.* (2021) **Empagliflozin in Heart Failure with a Preserved Ejection Fraction** *N Engl J Med* **385**:1451–1461
10. Meng Q., Lima J. A., Lai H., Vlahov D., Celentano D. D., Strathdee S., Nelson K. E., Tong W., Lai S (2002) **Use of HIV protease inhibitors is associated with left ventricular morphologic changes and diastolic dysfunction** *J Acquir Immune Defic Syndr* **30**:306–310
11. Schuster I. *et al.* (2008) **Subclinical cardiac abnormalities in human immunodeficiency virus-infected men receiving antiretroviral therapy** *The American journal of cardiology* **101**:1213–1217
12. Hsue P. Y., Hunt P. W., Ho J. E., Farah H. H., Schnell A., Hoh R., Martin J. N., Deeks S. G., Bolger A. F (2010) **Impact of HIV infection on diastolic function and left ventricular mass** *Circulation. Heart failure* **3**:132–139

13. Berzingi C., Chen F., Finkel M. S (2009) **p38 MAP kinase inhibitor prevents diastolic dysfunction in rats following HIV gp120 injection in vivo** *Cardiovascular toxicology* **9**:142–150
14. Kan H., Xie Z., Finkel M. S (2004) **p38 MAP kinase-mediated negative inotropic effect of HIV gp120 on cardiac myocytes** *American journal of physiology. Cell physiology* **286**:C1–7
15. Yuan Y., Kan H., Fang Q., Chen F., Finkel M. S (2008) **CXCR4 receptor antagonist blocks cardiac myocyte p38 MAP kinase phosphorylation by HIV gp120** *Cardiovascular toxicology* **8**:173–180
16. Kelly K. M. *et al.* (2014) **CCR5 inhibition prevents cardiac dysfunction in the SIV/macaque model of HIV** *Journal of the American Heart Association* **3**
17. Katsikis P. D., Mueller Y. M., Villinger F (2011) **The cytokine network of acute HIV infection: a promising target for vaccines and therapy to reduce viral set-point?** *PLoS pathogens* **7**
18. Keating S. M. *et al.* (2011) **The effect of HIV infection and HAART on inflammatory biomarkers in a population-based cohort of women** *Aids* **25**:1823–1832
19. Ramirez L. A., Arango T. A., Thompson E., Naji M., Tebas P., Boyer J. D (2014) **High IP-10 levels decrease T cell function in HIV-1-infected individuals on ART** *J Leukoc Biol* **96**:1055–1063
20. Roff S. R., Noon-Song E. N., Yamamoto J. K (2014) **The Significance of Interferon-gamma in HIV-1 Pathogenesis, Therapy, and Prophylaxis** *Front Immunol* **4**
21. Watanabe D. *et al.* (2019) **Clinical characteristics of HIV-1-infected patients with high levels of plasma interferon-gamma: a multicenter observational study** *BMC Infect Dis* **19**
22. Amirayan-Chevillard N., Tissot-Dupont H., Capo C., Brunet C., Dignat-George F., Obadia Y., Gallais H., Mege J. L (2000) **Impact of highly active anti-retroviral therapy (HAART) on cytokine production and monocyte subsets in HIV-infected patients** *Clinical and experimental immunology* **120**:107–112
23. Wada N. I., Bream J. H., Martinez-Maza O., Macatangay B., Galvin S. R., Margolick J. B., Jacobson L. P (2016) **Inflammatory Biomarkers and Mortality Risk Among HIV-Suppressed Men: A Multisite Prospective Cohort Study** *Clin Infect Dis* **63**:984–990
24. Bahrami H. *et al.* (2016) **Inflammatory Markers Associated With Subclinical Coronary Artery Disease: The Multicenter AIDS Cohort Study** *Journal of the American Heart Association*
25. Itzhaki I. *et al.* (2011) **Modelling the long QT syndrome with induced pluripotent stem cells** *Nature* **471**:225–229
26. Carvajal-Vergara X. *et al.* (2010) **Patient-specific induced pluripotent stem-cell-derived models of LEOPARD syndrome** *Nature* **465**:808–812
27. Yazawa M., Hsueh B., Jia X., Pasca A. M., Bernstein J. A., Hallmayer J., Dolmetsch R. E (2011) **Using induced pluripotent stem cells to investigate cardiac phenotypes in Timothy syndrome** *Nature* **471**:230–234
28. Sun N. *et al.* (2012) **Patient-specific induced pluripotent stem cells as a model for familial dilated cardiomyopathy** *Sci. Transl. Med* **4**

29. Lan F. *et al.* (2013) **Abnormal calcium handling properties underlie familial hypertrophic cardiomyopathy pathology in patient-specific induced pluripotent stem cells** *Cell Stem Cell* **12**:101–113
30. Liang P. *et al.* (2013) **Drug screening using a library of human induced pluripotent stem cell-derived cardiomyocytes reveals disease-specific patterns of cardiotoxicity** *Circulation* **127**:1677–1691
31. Navarrete E. G. *et al.* (2013) **Screening drug-induced arrhythmia [corrected] using human induced pluripotent stem cell-derived cardiomyocytes and low-impedance microelectrode arrays** *Circulation* **128**:S3–13
32. Sharma A. *et al.* (2014) **Human induced pluripotent stem cell-derived cardiomyocytes as an in vitro model for coxsackievirus B3-induced myocarditis and antiviral drug screening platform** *Circ Res* **115**:556–566
33. Burridge P. W., Holmstrom A., Wu J. C (2015) **Chemically Defined Culture and Cardiomyocyte Differentiation of Human Pluripotent Stem Cells** *Curr Protoc Hum Genet* **87**:21–21
34. Kumar A. A., Kelly D. P., Chirinos J. A (2019) **Mitochondrial Dysfunction in Heart Failure With Preserved Ejection Fraction** *Circulation* **139**:1435–1450
35. Droste J. A., Verweij-van Wissen C. P., Kearney B. P., Buffels R., Vanhorssen P. J., Hekster Y. A., Burger D. M. (2005) **Pharmacokinetic study of tenofovir disoproxil fumarate combined with rifampin in healthy volunteers** *Antimicrobial agents and chemotherapy* **49**:680–684
36. Kakuda T. N., Brochot A., Tomaka F. L., Vangeneugden T., Van De Castele T., Hoetelmans R. M. (2014) **Pharmacokinetics and pharmacodynamics of boosted once-daily darunavir** *The Journal of antimicrobial chemotherapy* **69**:2591–2605
37. Shiomi M., Matsuki S., Ikeda A., Ishikawa T., Nishino N., Kimura M., Kumagai Y., Irie S (2015) **Pharmacokinetic and bioequivalence evaluation of single-tablet and separate-tablet regimens for once-daily cobicistat-boosted elvitegravir in healthy Japanese male subjects: A randomized, two-way crossover study** *Clinical pharmacology in drug development* **4**:218–225
38. Wang L., Soon G. H., Seng K. Y., Li J., Lee E., Yong E. L., Goh B. C., Flexner C., Lee L (2011) **Pharmacokinetic modeling of plasma and intracellular concentrations of raltegravir in healthy volunteers** *Antimicrobial agents and chemotherapy* **55**:4090–4095
39. Castro L. R., Verde I., Cooper D. M., Fischmeister R (2006) **Cyclic guanosine monophosphate compartmentation in rat cardiac myocytes** *Circulation* **113**:2221–2228
40. Kiso H., Ohba T., Iino K., Sato K., Terata Y., Murakami M., Ono K., Watanabe H., Ito H (2013) **Sildenafil prevents the up-regulation of transient receptor potential canonical channels in the development of cardiomyocyte hypertrophy** *Biochemical and biophysical research communications* **436**:514–518
41. Lee D. I. *et al.* (2015) **Phosphodiesterase 9A controls nitric-oxide-independent cGMP and hypertrophic heart disease** *Nature* **519**:472–476
42. Reinke Y. *et al.* (2015) **The soluble guanylate cyclase stimulator riociguat and the soluble guanylate cyclase activator cinaciguat exert no direct effects on contractility and relaxation of cardiac myocytes from normal rats** *European journal of pharmacology* **767**:1–9

43. Zhang S., Rodriguez R., Scholz P. M., Weiss H. R (2006) **Functional interaction of a beta-adrenergic agonist and cyclic GMP phosphodiesterase inhibitor in control and hypertrophic cardiomyocytes** *Pharmacology* **76**:53–60
44. Arnold J. R. *et al.* (2022) **Prevalence and Prognostic Significance of Microvascular Dysfunction in Heart Failure With Preserved Ejection Fraction** *JACC Cardiovasc Imaging* **15**:1001–1011
45. Markley R. *et al.* (2023) **Abnormal left ventricular subendocardial perfusion and diastolic function in women with obesity and heart failure and preserved ejection fraction** *Int J Cardiovasc Imaging* **39**:811–819
46. Churko J. M., Burrridge P. W., Wu J. C (2013) **Generation of human iPSCs from human peripheral blood mononuclear cells using non-integrative Sendai virus in chemically defined conditions** *Methods in molecular biology* **1036**:81–88
47. Romero-Tejeda M., Fonoudi H., Weddle C. J., DeKeyser J. M., Lenny B., Fetterman K. A., Magdy T., Sapkota Y., Epting C. L., Burrridge P. W (2023) **A novel transcription factor combination for direct reprogramming to a spontaneously contracting human cardiomyocyte-like state** *Journal of molecular and cellular cardiology* **182**:30–43
48. Das A. *et al.* (2018) **Impairment of an Endothelial NAD(+)-H(2)S Signaling Network Is a Reversible Cause of Vascular Aging** *Cell* **173**:74–89
49. Lee D. C., Johnson N. P (2009) **Quantification of absolute myocardial blood flow by magnetic resonance perfusion imaging** *JACC Cardiovasc Imaging* **2**:761–770

Article and author information

Yuki Tatekoshi

Feinberg Cardiovascular and Renal Research Institute, and Northwestern University, Chicago, IL

Chunlei Chen

Feinberg Cardiovascular and Renal Research Institute, and Northwestern University, Chicago, IL

Jason Shapiro

Feinberg Cardiovascular and Renal Research Institute, and Northwestern University, Chicago, IL

ORCID iD: [0000-0003-0880-3142](https://orcid.org/0000-0003-0880-3142)

Hsiang-Chun Chang

Feinberg Cardiovascular and Renal Research Institute, and Northwestern University, Chicago, IL

Malorie Blancard

Department of Pharmacology, Northwestern University, Chicago, IL

Paul W. Burrridge

Department of Pharmacology, Northwestern University, Chicago, IL

Paul Burrridge

Department of Pharmacology, Northwestern University, Chicago, IL

Matthew Feinstein

Department of Medicine, Northwestern University, Chicago, IL

Richard D'Aquila

Department of Medicine, Northwestern University, Chicago, IL

Priscilla Hsue

Department of Medicine, University of California at San Francisco, San Francisco, CA

Hossein Ardehali

Feinberg Cardiovascular and Renal Research Institute, and Northwestern University, Chicago, IL, Department of Pharmacology, Northwestern University, Chicago, IL, Department of Medicine, Northwestern University, Chicago, IL

For correspondence: h-ardehali@northwestern.edu

ORCID iD: [0000-0002-7662-0551](https://orcid.org/0000-0002-7662-0551)

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Reviewer #1 (Public Review):

Summary:

This is an original manuscript submission by Tatekoshi et al entitled, "Human induced pluripotent stem cell-derived cardiomyocytes to study inflammation-induced diastolic dysfunction." Based on the premise that treated HIV individuals commonly have heart failure with preserved ejection fraction, yet robust animal models have not been established, the team developed iPS-CM models to study HFpEF with this angle in mind. The group established iPS-CMs using standard methods and studied TNF α and IFN γ effects on calcium transients. They observed that both cytokines increased calcium transient decay and downstroke times, which could be reversed by mitoTempo treatment in the case of TNF α . To determine how mitochondrial dysfunction may impact the cytokine-induced calcium transient changes, the team measured OCR treatment changes. They observed that NAC and TNF α co-treated cells demonstrated reduced OCR. The team went on to test the effects of antiretroviral therapies including tenofovir, raltegravir, elvitegravir, and darunavir at 3-10

uM levels in iPS-CMs. The team noted that ART treatments reversed the diastolic dysfunction associated with TNFa treatment suggesting that ART therapies may improve diastolic dysfunction that is associated with TNFa signaling directly in cardiomyocytes. Following up on this treatment effect, the team screened several other candidates across drug classes and identified that dapagliflozin (SGLT2i) reversed diastolic dysfunction induced by TNFa. Finally, the team collected human serum from patients with HIV+ patients from two hospitals - Northwestern with diastolic dysfunction by cMRI, and UCSF with normal diastolic function by echo. Both cohort serum samples did not change calcium transients in iPS-CMs. However, due to numerous and significant major methodological concerns, and the potential low impact of the study results, this manuscript is expected to be of very low impact to the field in its current form.

Strengths:

1. From a significance standpoint, understanding the mechanisms of HFpEF, particularly in conditions such as HIV would be very impactful.
2. Collecting HIV patient serum and identifying a plasma factor that impacts cardiac function could be very significant if successful.

Weaknesses:

- (1) I am not convinced how this study relates to HIV individual HFpEF, and the study design does not seem to be well thought out.
- (2) The connectivity of the study experiments is loose, and data analysis and conclusions are broadly overstated and misinterpreted.
- (3) For example the study lacks any measure of diastolic contractile function, and even if performed, the relevance of TNFa treatments to cells in vitro in these immature cell contexts would remain unclear. There is surprisingly no reported molecular analyses of potential mechanisms of the calcium transient changes. The study falls short in molecular detail and instead relies on drug treatments and responses that are hard to interpret with dosages that are not well justified and treatments that are numerous. Unclear what changes in calcium transients mean functionally without a comprehensive assessment of CM biomechanical contraction and relaxation measurements, and this would also require parallel molecular investigations of potential targets of any phenotypes observed.
- (4) Calcium transient data need to be better illustrated such as with representative peak tracings. The data overall is with too few samples, particularly given the inherent heterogeneity of iPS-CM studies. The iPS-CM system as a model for diastolic dysfunction remains unestablished.
- (5) There are unclear dose choices for the various ART drugs tested, as well as the other drugs tested such as SGLT2i. Besides the observation that SLC5A2 (SGLT2 target) is not established to be expressed in adult mammalian cardiomyocytes.
- (6) HIV plasma samples were not tested for cytokine levels, but this could be done to assess the validity of the final experiments. It is unclear what is being tested with these experiments.
- (7) The choice of serum controls from a second institution (UCSF) opens up concerns over batch effects unrelated to differences in diastolic dysfunction. However, there were no differences with the Northwestern samples. It is unclear why this data is included as it does not add to the impact of the study.

(8) There are concerns about the quality of the iPSC-CMs since there is no cell imaging or molecular analyses. Figure 5 Supplement 1 images are of low quality and low resolution to assess cell quality. Overall the iPSC-CM QC data is extremely sparse

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Reviewer #2 (Public Review):

The authors investigated the role of inflammatory molecules in diastolic dysfunction and screened antiviral and cardioprotective pharmacological agents for their potential to reverse inflammation-mediated diastolic dysfunction. This study focuses on heart failure with preserved ejection fraction (HFpEF) in people living with HIV (PLWH), a condition often challenging to study due to the lack of suitable animal models. Using human-induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs), researchers simulated HFpEF in vitro. They observed that inflammatory cytokines impaired cardiomyocyte relaxation, mimicking HFpEF, while SGLT2 inhibitors and mitochondrial antioxidants reversed this effect. Exposure to serum from HIV patients did not induce dysfunction in hiPSC-CMs. These findings suggest hiPSC-CMs as a promising model for understanding HFpEF mechanisms and testing potential treatments.

Comments to improve the study:

The manuscript is well-written, and the results are well-illustrated. However, there are some topics that are not well-connected, and the rationale and hypothesis are not clearly defined beforehand, such as mitochondrial membrane potential, mitochondrial ROS, and angiogenic potential.

As the hiPSC cardiomyocytes are treated with various reagents to measure diastolic dysfunction, it is important to confirm whether the treatment time and dose used were sufficient to exert a functional effect. Dose and time-dependent experiments are essential, or at least sufficient citations should be provided for selecting the dose for IFN and TNF.

After IFN and TNF treatment, determining the expression levels of molecular markers of DD/HFpEF is crucial. Again, if sufficient evidence is available, it can be cited.

The Methods section describes TMRE colocalization and immunofluorescence, but no images are provided.

The concentration of TNF and IFN in patients is critical, which was acknowledged and discussed as a limitation of the study by the authors. Authors should consider this aspect, and if not feasible, clinical reports should be cited to provide a rough estimation of their concentration.

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